

Tiger Strake

PROFILE

Stanford AeroAstro student focused on aerospace hardware, fixed-wing UAVs, high-power rockets, embedded controls, and fabrication-heavy test rigs.

EDUCATION

Stanford University

B.S. Aeronautics and Astronautics, expected 2029

Hochschule Hamm-Lippstadt Junior Campus

Dual-enrollment program, 2022-2024

Harvard University Summer School

Digital fabrication and planetary science, 2023

Evangelisches Gymnasium Lippstadt

2021-2025

AVIATION

- EASA PPL(A), SEP(L), night qualification
- FAA Private Pilot, ASEL, night operations
- Training: MEP, helicopter, high performance, tailwheel, complex

TECHNICAL STACK

CAD / analysis

Fusion 360, OpenRocket, Luminary CFD, SolidWorks, Onshape

Embedded

ESP32, FreeRTOS, Raspberry Pi, Python, telemetry, sensor logging

Fabrication

FDM, SLA, SLS, SLM, composites, CNC routing, vacuum bagging, laser cutting

Media

Final Cut, Premiere Pro, After Effects, Photoshop

ENGINEERING EXPERIENCE

Stanford Swift Solar Plane Project

2026-present

Engineering Team Member

- Contribute to a Stanford team building a solar-powered aircraft more than 5 m in wingspan for autonomous, multi-day flight.
- Support the long-term program goal of setting the world record for the longest continuous electric flight.

SkyRunners UAV, Stanford

2025-2026

Manufacturing and Structures Lead

- Led manufacturing and structures work for a Stanford fixed-wing UAV team, translating airframe design intent into buildable, serviceable parts.
- Validated a cleaner wing production workflow using a 4-axis hot-wire foam cutter after manual rib-and-foam cutting introduced too much asymmetry.
- Rebuilt the CAD model parametrically so chord, span, rib spacing, fuselage cross-section, and tail geometry update from a shared parameter table.

Fabrication Lab EG

2022-2024

Founder and Lab Lead · Evangelisches Gymnasium Lippstadt

- Founded and ran a school makerspace with five Bambu Lab X1Cs plus personal FDM, SLA, and laser-cutting equipment.
- Mentored 12 students across grades 6-10 in 3D printing, CAD, laser cutting, PCB design, and web development.
- Built a tiered machine-training process and handed the lab off to trained student successors before graduation.

European Space Agency (ESA)

Jul 2022

Intern, Darmstadt

- Worked around mission operations for BepiColombo and LISA Pathfinder, with a focus on attitude control systems and redundancy logic.
- Completed backup satellite system analysis and contributed recommendations to attitude-control documentation.

German Aerospace Center (DLR)

Apr-May 2022

Intern, Cologne

- Worked in a hypersonic wind-tunnel environment and analyzed airfoil behavior at Mach 10 under supervised facility protocols.
- Documented scramjet inlet and flow-path efficiency work from facility data and technical briefings.

European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT)

Mar-Apr 2022

Intern, Darmstadt

- Developed a space-systems education program covering satellite construction, rocketry, and orbital mechanics.
- Prepared, recorded, edited, and published a podcast interview with the Director General on low Earth orbit satellite operations.

CERTIFICATIONS

- MINT-EC Scholar with distinction
- EASA PPL(A), SEP(L)
- FAA Private Pilot, ASEL

LANGUAGES

- English - fluent
- German - fluent
- French - conversational
- Spanish - conversational

ADDITIONAL EXPERIENCE

Autohaus Hecker

Content producer, 2019-present.
Produce and edit social media content for a commercial truck dealership.

CURRENT FOCUS

Stanford Swift solar aircraft, L2 rocket build, Doorboard, ESP32 control systems, and flight training.

SELECTED PROJECTS

L1 and L2 High-Power Rocketry

2025-present

Builder / Systems Lead

- Built and bench-checked an L1 certification rocket; certification flight pending an insured NAR launch window.
- Started L2 vehicle build with fiberglass airframe planning, dual-deployment recovery, redundant altimeters, and independent battery/arming systems.

Stratosphere Balloon Payload

2022

Co-lead Engineer

- Built a 4-student science payload flown to approximately 30 km with a Raspberry Pi 2B, GoPro, Pi camera, GPS tracking, and sensor logging.
- Recovered full altitude, pressure, interior/exterior temperature, magnetic-field, and acceleration data from the complete flight profile.

Rocket Thrust Vector Control

2026

Mechanical and Embedded Integrator

- Built an ESP32-based TVC prototype for small rocket stabilization experiments, prioritizing hardware-in-loop validation before propulsion integration.
- Prototyped a compliant gimbal, characterized backlash on a bench fixture, and reworked linkage geometry after measured backlash exceeded predictions.

ESP32 Smart Light Switch

2026

Designer and Builder

- Built a self-contained servo retrofit for a wall switch; ESP32 hosts its own WiFi access point and local scheduling UI.
- Implemented FreeRTOS servo/web-server tasks and NVS-backed scheduling; web UI loads in under 200 ms on the local network.

Pulse Jet Test Rig

2025-present

Lead Builder and Test Engineer

- Designed a modular valveless pulse-jet fixture around repeatable geometry changes, fixed ignition routing, and defined instrumentation reference points.
- Completed initial chamber fabrication and support fixture; hot-fire testing remains gated on safety protocol and instrumentation review.

Additive Manufacturing Competition

2023

Individual Competitor

- Competed solo against company-sponsored teams across SLA, SLS, SLM, and FDM challenge rounds.
- Won most challenge rounds by adapting geometry and tolerance strategy to each additive process and functional part requirement.

Small Robotic Arm

2026

Designer and Builder

- Built a 3-DOF plus gripper desktop arm from printed PLA structure, ESP32 control, servo actuation, and a serial trajectory interface.
- Completed structure and forward-kinematics verification; further control work is currently paused.